Beyond Linsker's Infomax Principle: A Rate–Distortion Perspective on InfoNCE

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***Abstract*—InfoNCE is usually explained through mutual information, but with finitely many negatives its lower bound saturates and does not answer a more basic question: which information-theoretic problem does InfoNCE actually solve? We show that the answer is a rate–distortion problem that keeps the output distribution fixed. The score learned by InfoNCE acts as a distortion measure, and the process generating positive pairs is the unique optimal channel for the resulting compression problem. When both views are produced by the same augmentation process, this becomes marginal-preserving rate–distortion. Experiments recover arbitrary positive-pair channels, reveal a coarse-to-fine compression hierarchy in Markov dynamics, and show on CIFAR-100 that the learned distortion preserves semantic information more efficiently than pixel distance or an untrained encoder.**

I. Introduction

InfoNCE is a central objective in self-supervised representation learning [1], [3], [4]. Its usual information-theoretic explanation invokes Linsker's Infomax principle [5]: contrastive learning is viewed as preserving mutual information between two views [1], [2]. This explanation is incomplete at finite K. If $L_K^\star \triangleq \inf_d L_K(d, p_{Y|X}, \tau)$, then

$$\log(K+1) - L_K^\star \le \min\{\log(K+1), I(X;Y)\}, \tag{1}$$

so the optimized value cannot distinguish mutual informations above $\log(K+1)$. This has motivated explanations based on downstream utility [8], hyperspherical geometry [9], representation identifiability [10], and density-ratio estimation [11]. What remains is an operational information-theoretic problem that shares the finite-K optimal channel and critic.

Our answer is *fixed-output rate–distortion* (RD). InfoNCE observes the positive-pair channel $W^\star = P_{Y|X}$ and identifies, up to its natural output-dependent gauge, an effective distortion $d^\star$. If the RD problem fixes the output marginal to $\mu = P_Y^\star$, then the same $W^\star$ is its unique optimizer, and the optimality gap is an exact joint KL divergence. Thus InfoNCE runs the inverse direction of RD:

$$W^\star \xrightarrow{\text{InfoNCE}} [d^\star] \xrightarrow{\text{fixed-output RD}} W^\star. \tag{2}$$

The output constraint is not added merely to force a desired conclusion. It is exactly what makes the output-dependent ambiguity of the InfoNCE critic invisible on the RD side.

When an underlying sample U is passed twice through the same augmentation channel, $X, Y \mid U \overset{\text{i.i.d.}}{\sim} Q(\cdot \mid U)$, the two view marginals agree. Fixed-output RD then becomes marginal-preserving RD (MP–RD). We also give a secondary converse: for an exact symmetric critic that is optimal in both contrastive directions, an injective Gibbs kernel forces the marginals to match.

Classical RD provides an operational theory of lossy compression [6], [7]. Fixed-output coding has been studied as distribution-preserving quantization and output-constrained source coding [12]–[14], and is related to rate–distortion–perception theory [15], [16]. Its Lagrangian is an entropic transport or static Schrödinger problem computable by Sinkhorn scaling [17], [18], while inverse optimality is related to probabilistic source–channel matching [19]. Our contribution is the exact finite-K bridge to InfoNCE, including its gauge and KL-gap structure, together with evidence that the induced distortion is meaningful in learned representations.

II. Setup

We state the exact result for finite alphabets and full-support channels. Draw J uniformly from $\{0, \dots, K\}$, draw $X_0, \dots, X_K \overset{\text{i.i.d.}}{\sim} p_X$, and then draw $Y \sim W(\cdot \mid X_J)$. For $K \ge 1$ and $\tau > 0$, define

$$L_K(d, W, \tau) \triangleq -\mathbb{E} \log \frac{e^{-d(X_J, Y)/\tau}}{\sum_{i=0}^K e^{-d(X_i, Y)/\tau}}. \tag{3}$$

Lemma 1 (Bayes-optimal finite-K critic [11]): A score d minimizes (3) if and only if

$$-\frac{d(x,y)}{\tau} = \log \frac{W(y \mid x)}{p_Y(y)} + a(y) \tag{4}$$

for some function a. Hence the optimal critic class is the same for every finite $K \ge 1$.

The function $a(y)$ cancels because the same Y appears in every candidate logit. Equivalently, population cross-entropy is conditional entropy plus the expected KL divergence from the Bayes candidate posterior.

For a prescribed output law μ, let

$$\mathcal{W}_\mu(p_X) \triangleq \{V_{Y|X} : P_Y^V = \mu\}, \tag{5}$$

and define

$$R_\mu(D; d) \triangleq \inf_{\substack{V\in\mathcal{W}_\mu(p_X):\\ \mathbb{E}_V d\le D}} I_V(X;Y). \tag{6}$$

The MP–RD problem is $R_{p_X}(D; d)$. Since both marginals are fixed, replacing d by

$$\widetilde{d}(x,y) = \alpha d(x,y) + u(x) + v(y), \qquad \alpha > 0, \tag{7}$$

only shifts and rescales the distortion level and leaves the optimizing channel unchanged.

III. The Fixed-Output RD Bridge

Theorem 1 (Finite-K InfoNCE and fixed-output RD): Let $W^\star \in \mathcal{W}_\mu(p_X)$ have full support and let $d^\star$ be Bayes-optimal for (3). Define

$$\mathcal{J}_{d^\star}(V) \triangleq I_V(X;Y) + \frac{1}{\tau}\mathbb{E}_V d^\star(X,Y). \tag{8}$$

Then every $V \in \mathcal{W}_\mu(p_X)$ satisfies

$$\mathcal{J}_{d^\star}(V) - \mathcal{J}_{d^\star}(W^\star) = D_{\mathrm{KL}}(P_X V \,\|\, P_X W^\star). \tag{9}$$

Consequently, $W^\star$ is the unique minimizer of $\mathcal{J}_{d^\star}$ and solves $R_\mu(D^\star; d^\star)$ at $D^\star = \mathbb{E}_{W^\star} d^\star$.

Conversely, if $W^\star$ is a full-support minimizer of $I_V + \beta\mathbb{E}_V\delta$ over $\mathcal{W}_\mu(p_X)$ for some $\beta > 0$, then an RD-gauge representative of δ is Bayes-optimal for finite-K InfoNCE with channel $W^\star$.

Proof: Lemma 1 gives $-d^\star/\tau = \log[W^\star(y \mid x)/\mu(y)] + a(y)$. Hence, for $V \in \mathcal{W}_\mu(p_X)$,

$$\begin{aligned}\mathcal{J}_{d^\star}(V) &= \mathbb{E}_{P_X V} \log\frac{V(Y\mid X)}{W^\star(Y\mid X)} - \mathbb{E}_\mu a(Y)\\ &= D_{\mathrm{KL}}(P_X V \| P_X W^\star) - \mathbb{E}_\mu a(Y),\end{aligned} \tag{10}$$

which proves (9); the constrained claim follows by the Lagrange-optimality inequality. Conversely, the fixed-output KKT conditions give functions u, v such that

$$\log\frac{W^\star(y\mid x)}{\mu(y)} = -\beta\delta(x,y) - u(x) - v(y). \tag{11}$$

Then $\bar{d} = \tau\beta\delta + \tau u$ is an RD-gauge representative and satisfies $-\bar{d}/\tau = \log(W^\star/\mu) + v(y)$, so Lemma 1 applies. $\square$

Corollary 1 (Exchangeable views): If $X, Y \mid U \overset{\text{i.i.d.}}{\sim} Q(\cdot \mid U)$, then $p_X = p_Y$. Theorem 1 therefore specializes to MP–RD: InfoNCE recovers an operational distortion for which the augmentation channel is the unique marginal-preserving RD optimizer.

Ordinary one-direction InfoNCE does not require $p_Y = p_X$; it always produces the fixed-output problem with $\mu = p_Y$. The matched-view assumption selects MP–RD from this broader family. The distinction is operationally important: only with fixed $P_Y^V = \mu$ does every InfoNCE gauge representative $d(x,y) + g(y)$ define the same RD problem and yield the joint-KL identity (9).

A. A structural converse

Let the reverse InfoNCE loss swap the anchor and candidate variables while using the same score. Call d an exact symmetric bidirectional realization if $d(x,y) = d(y,x)$ and it is Bayes-optimal in both directions. Define $G_d(x,y) = e^{-d(x,y)/\tau}$, and call G_d distribution-injective if $G_d p = G_d q$ for probability vectors p, q implies $p = q$.

Proposition 1 (When matched marginals are forced): If d is an exact symmetric bidirectional realization, the positive-pair support is connected, and G_d is distribution-injective, then $p_X = p_Y$.

Proof: Forward and reverse optimality give $-d/\tau = \imath(x;y) + a(y) = \imath(x;y) + b(x)$. Connected support makes a, b one constant. Marginalizing the resulting joint law gives $G_d p_Y = c\mathbf{1}$ and $G_d^\mathsf{T} p_X = c\mathbf{1}$. Symmetry and injectivity imply $p_X = p_Y$. $\square$

For $d_f(x,y) = \|f(x) - f(y)\|_2^2$, G_d is Gaussian on the embedding points. Pairwise distinct finite embeddings give a nonsingular matrix [20]; if f intentionally merges inputs, the conclusion weakens to equality of representation marginals $f_\# p_X = f_\# p_Y$.

IV. Experiments

We use three complementary experiments: a direct test of Theorem 1, a controlled demonstration of the induced compression hierarchy, and a practical semantic test on learned image representations.

A. Arbitrary positive-pair channels

We draw random positive matrices and Sinkhorn-scale them to prescribed marginals, producing generic channels that are not generated from a known distortion. We consider both $p_Y = p_X$ and $p_Y \neq p_X$, train tabular InfoNCE critics for $K \in \{1, 4, 32\}$, and reconstruct the channel by Sinkhorn-scaling the learned Gibbs kernel to (p_X, p_Y). We use ten independently generated couplings and a multistep learning-rate schedule; the horizontal axis in Fig. 1(a) counts positive pairs processed.

At 9.2×10^5 pairs, median row KL for $K = 32$ is 5.48×10^{-4} in the matched case and 5.62×10^{-4} in the unmatched case. Thus the recovery is not

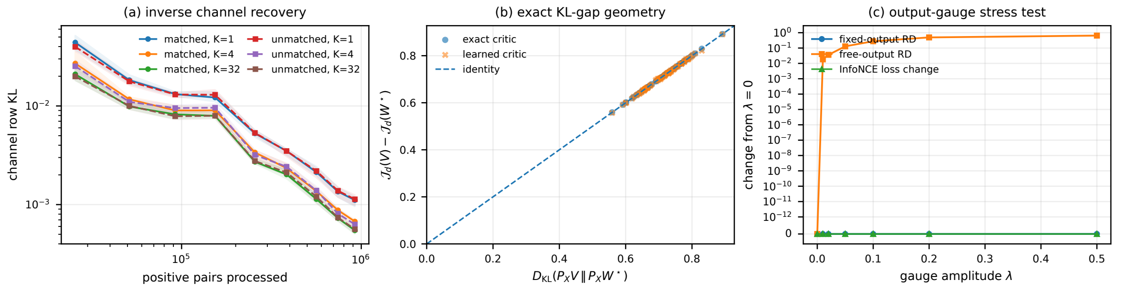

Fig. 1. Direct test of the fixed-output bridge. (a) Starting from arbitrary full-support couplings, finite-K InfoNCE followed by fixed-output Sinkhorn scaling recovers both matched and unmatched positive-pair channels; curves show medians and 10–90% ranges over ten couplings. (b) The exact critic satisfies the KL-gap identity, and a learned critic closely follows it. (c) Adding $\lambda g(y)$ changes neither InfoNCE nor fixed-output RD, while the free-output RD solution changes substantially.

specific to MP–RD; matched and unmatched channels behave nearly identically, as predicted by the fixed-output theorem. All finite K values recover the same population target, although larger K gives lower finite-training error at a fixed positive-pair count.

Figure 1(b) probes the stronger objective statement. For random competing couplings with the same marginals, the exact critic satisfies (9) to numerical precision. A representative learned $K = 32$ critic has channel row KL 5.17×10^{-4} and mean absolute KL-gap error 2.01×10^{-3} nats. Finally, Fig. 1(c) isolates why the output constraint is the clean counterpart of InfoNCE: adding an arbitrary $\lambda g(y)$ leaves the InfoNCE loss and fixed-output RD channel unchanged to numerical precision, while the free-output RD channel moves by up to 0.65 in weighted total variation.

B. Hierarchical Markov coarse-graining

We construct stationary reversible chains with 64 states arranged into four macro-communities, each containing four micro-communities of four states. Positive pairs are adjacent states and negatives follow the stationary marginal. After learning a tabular critic, we solve the induced MP–RD problem over a range of rates. Figures 2(a)–(b) show that the learned channel recovers both the nested block structure and the stronger within-state transition pattern. To avoid double counting, Fig. 2(c) reports normalized macro information, the additional micro-level information beyond the macro label, and the remaining state information beyond the micro label.

Across ten independently generated chains, the recovered transition channel has median row KL 9.71×10^{-3}. The learned hierarchy closely matches the oracle: the median rates required to retain 90% of macro-, incremental micro-, and incremental state-level information are 1.788, 3.002, and 4.020 nats. At rate $\log 4$, it retains 76.5% of macro information but only 21.1% of the incremental micro information and 2.4% of within-micro state information. At the same rate, Hamming and random spherical distances retain only 22.1% and 10.4% of macro information [Fig. 2(d)]. Thus the learned critic specifies a controlled coarse-to-fine compression hierarchy rather than merely reconstructing the transition classifier.

C. Semantic compression on CIFAR-100

We train a SimCLR-style ResNet-18 [4] for 400 epochs on CIFAR-100 [21], using no labels. The theorem-facing distortion is squared distance in the normalized projection space z used by the InfoNCE loss. On a balanced held-out set of 2,000 images, with uniform empirical marginal u, we solve

$$\pi_\beta \in \arg\min_{\pi\in\Pi(u,u)} D_{\mathrm{KL}}(\pi\|u\otimes u) + \beta\mathbb{E}_\pi d(X,\widetilde{X}) \tag{12}$$

by log-domain Sinkhorn scaling. CIFAR-100 labels are used only afterward to measure the normalized mutual information retained between the original and reconstructed superclass or fine class. We compare the InfoNCE distortion with pixel MSE, an architecture-matched untrained encoder, the pre-projection SimCLR representation h, and a supervised reference.

Figure 3 shows broad semantic-rate gains rather than a single selected operating point. At rate 4 nats, the InfoNCE critic retains 45.9% of superclass information and 47.0% of fine-label information, compared with 32.4% and 37.7% for pixels. To retain 80%, the critic requires 6.27 nats for both label levels, versus 6.80 and 6.62 nats for pixels and nearly identical rates for the random encoder. Results are stable across three independently trained models. The pre-projection representation h is slightly more semantically efficient than z, which is consistent with standard SimCLR practice; the important point here is that z, the space defining

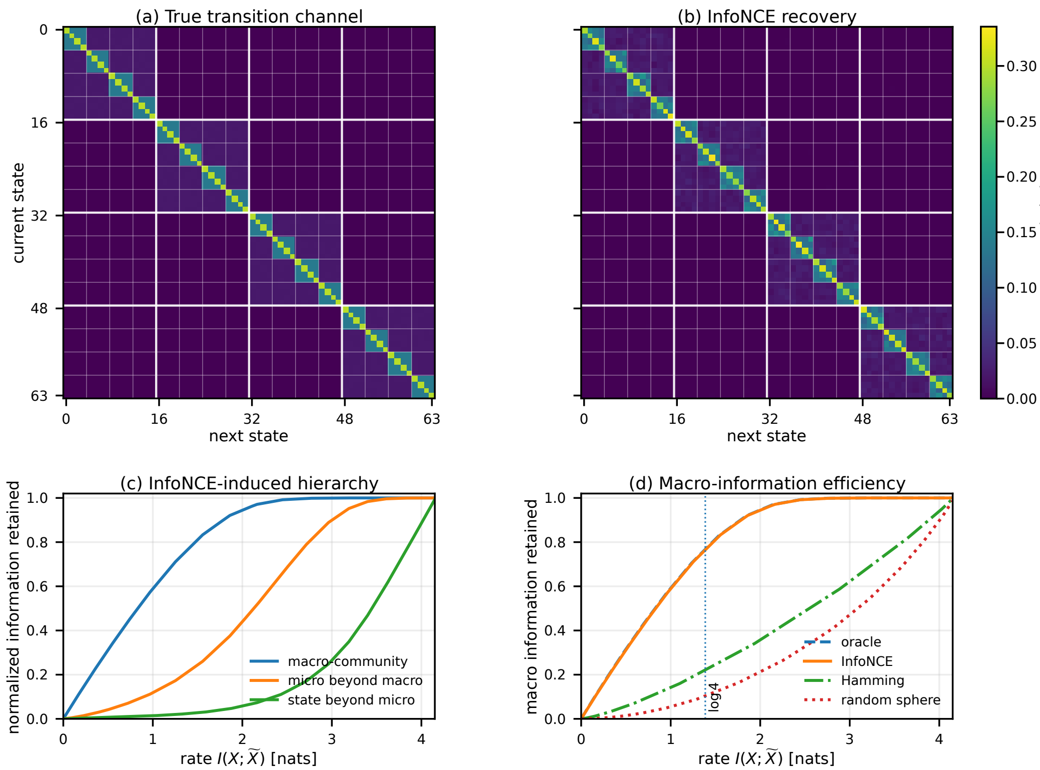

Fig. 2. Hierarchical Markov coarse-graining. Ten reversible 64-state chains have four macro-communities, each divided into four micro-communities. (a)–(b) A representative true transition channel and its InfoNCE recovery; white lines mark the nested micro- and macro-community blocks. (c) The induced MP–RD curve preserves macro information first, then incremental micro information, and finally within-micro state identity. (d) Its macro-information curve nearly matches the oracle and is substantially more rate-efficient than Hamming or random spherical geometry. Curves are medians over ten independently generated chains; the representative recovery has row KL 9.57×10^{-3}.

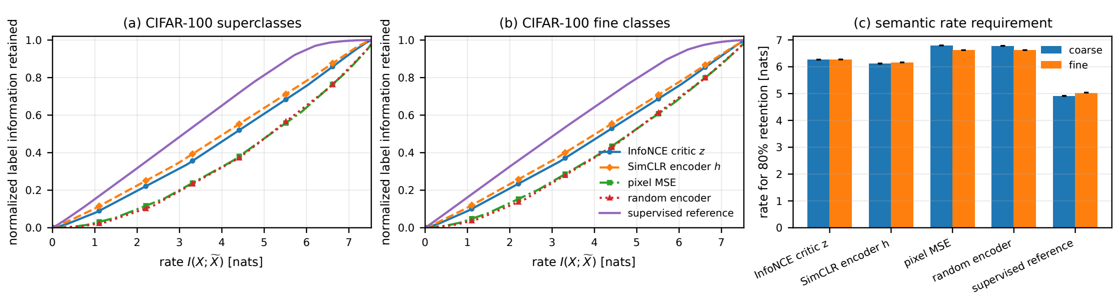

Fig. 3. Semantic compression on CIFAR-100. Squared distance in the normalized InfoNCE projection space preserves superclass and fine-label information at lower information rates than pixel MSE or an untrained encoder. Curves and error bars summarize three training seeds; labels are used only for evaluation. The pre-projection SimCLR representation h is slightly stronger, while the supervised model is an expected reference ceiling.

the actual InfoNCE critic, already yields a substantially better operational distortion than nonsemantic baselines.

The superclass and fine-label curves are close, so we do not claim a sharp coarse-to-fine phase transition. The supported practical conclusion is narrower and directly relevant to the theorem: the critic learned by InfoNCE allocates a given information budget to semantic distinctions more efficiently than pixel or random geometry.

V. Discussion

A. Why the output constraint is the right counterpart

Theorem 1 is stronger than the observation that density ratios and RD optimizers both admit Gibbs forms. InfoNCE determines only an equivalence class

$$d(x, y) \sim d(x, y) + g(y), \tag{13}$$

because $g(Y)$ cancels from every candidate softmax. If the RD output law is fixed, $\mathbb{E}_V g(Y) = \mathbb{E}_\mu g(Y)$ for every feasible V, so precisely the same equivalence class defines one optimization problem. This alignment yields both the exact KL gap and uniqueness. With a free output marginal, one may choose a canonical critic representative and construct an ordinary RD problem, but changing g generally changes the optimizer. Figure 1(c) isolates this distinction experimentally.

The condition $p_Y = p_X$ should therefore be interpreted at the correct level. It is not required for ordinary one-direction InfoNCE or for the general fixed-output bridge. It becomes relevant when the two views have the same law, in which case the operational output constraint is exactly the marginal-preserving one. Exchangeable augmentation provides this equality directly. Proposition 1 gives a separate converse under stronger geometric assumptions: an exact symmetric bidirectional critic with an injective kernel cannot support unmatched marginals.

B. From a critic to a compression hierarchy

The fixed-output Lagrangian also gives a practical procedure. Writing $\pi(dx, dy) = p_X(dx)V(dy \mid x)$, for any multiplier $\beta > 0$ we solve

$$\min_{\pi \in \Pi(p_X, \mu)} D_{\mathrm{KL}}(\pi \| p_X \otimes \mu) + \beta \mathbb{E}_\pi d. \tag{14}$$

If $R_\beta(dx, dy) \propto e^{-\beta d(x,y)} p_X(dx)\mu(dy)$, then (14) is the static Schrödinger projection $\min_{\pi \in \Pi(p_X, \mu)} D_{\mathrm{KL}}(\pi \| R_\beta)$ and can be computed by Sinkhorn scaling. Sweeping β turns a critic into an entire hierarchy of compressed couplings. The generic experiment uses this projection to recover the original channel; the Markov experiment exposes an explicit coarse-to-fine hierarchy; and CIFAR asks which semantic distinctions survive at reduced rate.

This gives a concrete workflow from positive pairs to an InfoNCE critic and then to a fixed-output RD curve. The output law encodes the operational requirement: $\mu = p_X$ for exchangeable views, or another prescribed law for constrained reconstruction. The theory therefore identifies both the learned fidelity criterion and the constraint that makes it operationally well-defined.

C. Scope and limitations

The exact theorem concerns population risk, full-support finite channels, and negatives drawn from the corresponding source marginal. Finite batches, restricted neural critics, mismatched negative sampling, and incomplete optimization introduce approximation error. A normalized neural encoder restricts the critic to a low-dimensional spherical geometry, so practical InfoNCE should be viewed as projecting the inverse RD distortion into that family. Kernel noninjectivity may represent either desired invariance or pathological collapse; the distinction depends on whether the collapsed directions are irrelevant to the intended fidelity criterion.

The experiments have different roles. The generic study directly tests the theorem because the coupling is known and the critic unrestricted. The hierarchical Markov model gives controlled ground truth for the induced compression order, while CIFAR-100 tests whether a learned neural critic yields useful empirical MP–RD geometry. CIFAR's gain over pixel and random baselines supports that interpretation, although its close superclass and fine-class curves do not establish a universal coarse-to-fine phase transition.

VI. Conclusion

Finite-K InfoNCE is an inverse fixed-output RD procedure. It learns an operational distortion for which the observed positive-pair channel is the unique optimizer, and the full fixed-output objective gap is a joint KL divergence. Exchangeable views specialize the result to MP–RD, while symmetric bidirectional realizability and kernel injectivity explain when matched marginals are forced by the geometry itself. The experiments verify the exact bridge, recover a controlled Markov compression hierarchy, and show that the same operational view yields semantic rate gains in a practical learned representation.

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